IoT and Cloud Based Healthcare Solution for Diabetic Foot Ulcer

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***Abstract*—Iot plays a vital role in solving real time problems in the field of Healthcare. Abundant problems can be rectified with optimal use of IOT Healthcare. It can be applied to detect Diabetes at early stages, detection of Foot ulcers, anomaly in heart rate and similar scenario. The paper proposed the plan and its working in Healthcare using IOT to detect ulcer in the foot of diabetic patients. The given model examines the medical condition of ulcer cause by diabetic and notify in case of aberration. Node MCU development board plays a vital role in its model development and stores and tracks the medical report of the Patient. It also helps in real time sharing of large chunks of data with great efficiency. Indeed this model slack off major time consuming efforts like regular visits to doctor and provide real time update with regards to patient.**

Keywords—cloud Computing. Internet of Things(IoT),Sensor, diabetic foot Ulcer, Monitoring , Alerts System

I. INTRODUCTION

Internet of things- Connecting everything with internet for the effective, multipurpose, time saving process. Here thing stands for each possible living or non-living which can be connected through internet to process data and transmit it to receiver. Researcher foreseen the usability of IOT in our day-to-day life. The smartwatch, which comprises of n number of activities from basic watch to calculating heartbeats everything processed in such a small device. This is an example of IOT. Another major example, which we can observe in this era, is "Smart Cities" by definition it stands a city, which is self-reliance, self-sustainable and independent of other factors can be monitored, analyzed and be processed to give fortuitous outcomes.

Integration of IOT with Healthcare has been a major outbreak in technological advancement for a better and prosperous human life. The technological advancement not only filled the vast gap between technology and health but also provided profound results. The very important example is emergence of platforms for IOT in healthcare like various smartwatches, which can easily track the heartbeat, blood pressure, oxygen leveler etc. All things at one place connected with a server to exchange data and compare it with the predefined situations to provide accurate results and recommendation accordingly. The healthcare sector has seen surge of such IOT platforms, which eases continues time devotion and disguised unemployment. Various patient's healthcare reports be tracked down easily within seconds, the cloud computation provides an abundant details which need to be looked into before proceeding. The data stored previously can be access through IOT in healthcare.

Diabetes mellitus is the major reason of high blood sugar in human body which results in various disorders in human body. High blood sugar may result in low immune system, liver damage and many other organ failures. Diabetic foot ulcer is one the disease which occurs due to diabetes. Every year 20% of the diabetic patients are hospitalized because of foot ulcer. In this disease it has been found that the limbs amputation is higher than an normal person. This may result in infection and low healing behavior. In majority of cases, the cause for the foot ulcer is altered architecture of the foot due to neuropathy resulting ion abnormal pressure points.

In this work we have proposed a cloud based IoT solution for foot ulcer prediction. The work will be able to identify abnormality in the patient behavior and can send alert to the doctor and patient. The work is divided into five sections Introduction, related work, proposed work, results and conclusion.

II. RELATED WORK

In COVID pendamic setuatino IoT and health care systems has proved to be very helpful to diagnose the health of a patient remotely. These systems are very beneficial to the world also.

In [1] 1.S. Amendola et.al. has proposed an IoT solution for healthcare using RFID. The research work titled RFID Technology for IoT-Based Personal Healthcare in Smart Spaces proposed a model to revolutionizing the traditional participatory medical models can be advances with the coagulation of IOT sensor spread with the environment with sole purpose to monitor the user's health issues, and refer them remotely for assistance. It is a working model with low cost, energy consumption, and disposable sensors. This coagulation of devices lead to survey through the recipient's body to gather information.In [2] A. Ukil et.al. has showcased the importance of existing IoT healthcare systems. The work proposed the importance of anomaly detection discuss about the use of IoT revolutionizing the concept of traditional methods. With few sensors and devices the anomaly in human body can be detected easily, the author is mainly concerned about the cardiac disease

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prevention with promising results. The anomaly is detected with low false negative alarm. In [3] S. Tyagi has proposed an healthcare solution using IoT and cloud for better healthcare services. The work proposed the impact of IoT in day-to-day life from various field of interest and addresses some essential parameters involved in application of IoT. The paper make it pragmatic and examined the role of IoT in healthcare delivery with the help of cloud based conceptual framework benefitting it.

In [4] Mahmud.R et.al. has proposed the challenges faced in healthcare using cloud computing to generate favorable and actual result, due to multi-hop distance from the economic aspects, geographical centralized architecture, data source. The introduction of Fog based IoT Healthcare solutions opens up the opportunity to integrate cloud fog services, generating a solution to traditional long due problem of cloud

In [5] Kim et.al. has presented the role of IoT in modernizing the traditional methods of analyzing the healthcare of a person by various critical factors such as profession of service provider, task scope, devices, expert support and many more. This data and medical history was examined within 435 respondents. The analysis of data and different medical histories were done accordingly. The results depicted the use of safe and trustworthy healthcare among the general respondents where as patient with medical history had significant amount of influence.

III. PROPOSED MODEL

We have proposed a monitoring system for diabetic patients health which is astute to retrieve and store the data from the system, in our case it is patient

The brief about the proposed model and system architecture propound for ulcer in foot of diabetics patients using internet of things and retrieved data is further analyzed by perpetuate logging of following parameters 1) Heartbeat 2) Body temperature 3) Dynamic foot pressure 4) Blood Pressure. The analytical comparison will help doctor and patient to analyze and predict the symptoms of foot ulcer, if required will alert the patient's trusted doctor with existing state along with medical history. Foot ulcer is major issue of concern leading to hospitalization. A paper published by Milkha D Madhale suggested " The analytical study of foot pressure point in diabetic patients when compared a normal person resulted in variation in pressure according to age"

The main purpose is to develop a Diabetic Foot Ulcer detection kit by optimally using flexi force sensor of unit upto 100 lbs and consisting of 6 pressure points. It examines the data 6 times a day whenever the patient is right position, ie : Standing straight, along with it the heartbeat, blood pressure and body temperature is also measured. These factors also contributes in complications faced by diabetic patients . All these readings are transmitted using Bluetooth connection with patients samrt phone and collected in the cloud database for further use.

A. Hardware

The IoT module in which our model is comprised is the Node MCU development board. It is an open source IoT platform, which includes the firmware which operates on ESP8266 Wi-Fi SOC from Espressif Systems, integrated with wifi and bluetooth and hardware which rely on ESP-12 module. NodeMCU ESP8266 have tensilica 32 bit RISC CPU Xtensa LX106, supported by RTOS to operate with in band of 80 to 160 MHz adjustable clock frequency. Second, Flexi sensor used is a two terminal device with non polarized ends (No positive no nagative) attached with patients foot to measure the foot points pressure. Third, the heart beat sensor(XD58C pulse sensor) which register the heart beats per second. Fourth DHT11 which is a temperature sensor, registering temperature of the patient. Blood pressure sensor is used to denote the blood pressure of the patient recorded over a period of time. All data is stored in the cloud storage for analysis over the period to determine any aberrant in patients data.

The analytical comparison of the received data of patient from the hardware with the data of normal person, if found any anomaly, it would notify the trusted doctor and patient through web app

The system is divided into three phases:

1. Sensing
2. Retrieving & analysis
3. Notify and reports

Sensing

The very first phase of the process is sensing, with the help of different devices various parameters are collected over the NodeMCU board and is further transmitted to cloud storage using internet. Further process are done in the cloud.

Retrieving & Analysis

In this phase the data from all different devices, 3 times a day, collected in the cloud from various devices is coagulated to check the health of the patient. In this phase only the analysis of the data and finding anomaly in data with comparison to normal person's recorded figures. Figure 1(a), (b) depicts the relation of patients age and diabetes which may cause Ulcer. The cloud is well established to represent an analytical study based on the parameters like blood pressure and foot pressure with an healthy person so that system can raise a notification of anomaly

Notify & reporting

In this phase the anomaly, which can be either raise in blood pressure or foot pressure are recorded in form of analytical report which enlightens the issue, notify so that quick action should be taken against it to revent future health degradation. Such reports are very helpful to make precursor move against the odds. As sharing this report weekly maintains transparency between patient and doctor. Helping doctor to analyse situation far better then ever, enabling a change in medication and living conditions

- The hardware placed under the foot of patient, will retrieve the data 3 times a day from 6 pressure points.
- Then this data is stored in the cloud where it is analyzed with the preexisting data of normal person.
- If the anomaly is found then an alert is engender from the web application to both the trusted doctor enlisted in the app and the patient.

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- The device store the data for further use

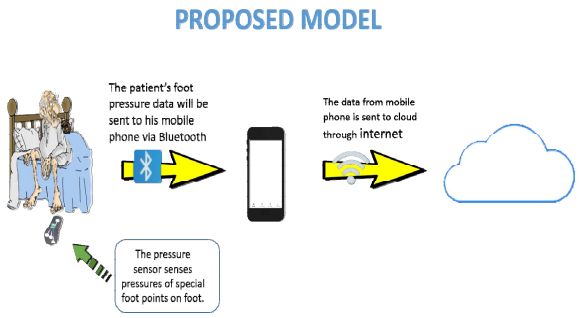

Fig. 1. Data Collection

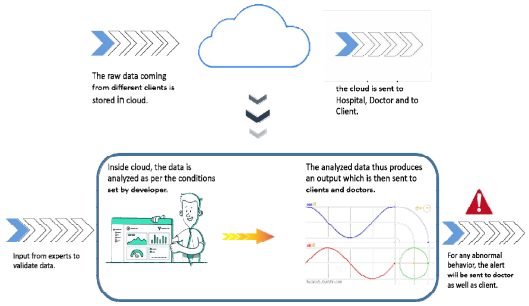

Fig. 2. Data Analysis and Alert

Fig. 3. Health reporting and messaging model

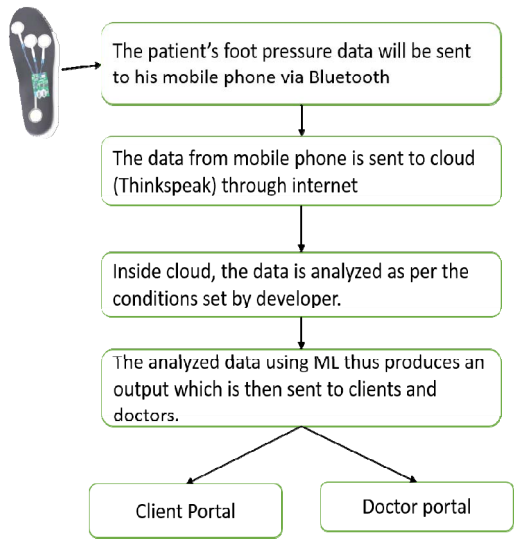

Fig. 4. Flow diagram.

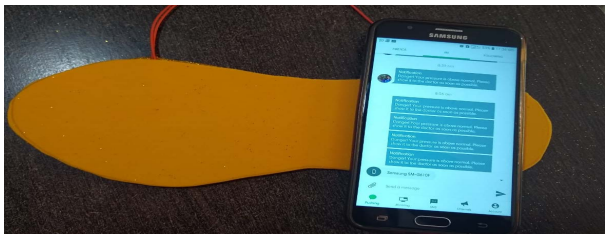

Fig. 5. Final product

IV. Results

This section demonstrates the results and output of the cloud readings and the IoT device readings. The solution is able to detect the abnormality based on the ruleset designed in the cloud platform. The cloud platform then sends the message to the doctor with the live data and type of abnormality in health.

Sensing

6-foot pressure (volts) points along temperature(Fahrenheit) are taken to predict DFU's complications.

TABLE I. Data Sensing

Sensors (V/F)	Test case 01	Test case 02	Test case 03
Sensor1	97.6752	94.7865	95.7890
Sensor2	102.6725	99.7865	100.7890
Sensor3	92.6752	89.7865	90.7890
Sensor4	106.6580	103.7865	104.7865
Sensor5	85.9865	82.9865	83.5643
Sensor6	92.1435	89.7865	90.5643
Temperature Sensor	63.8900	61.7463	69.5643

Harvesting and analyzation

The data of pressure sensor is being uploaded to ThingSpeak cloud platform and is being analyzed using push bullet where patient and diagnostics can visualize their previous records. Figure 4 a, b, c, d, e & f shows the sensed data of 6 pressure sensors data over cloud platform.

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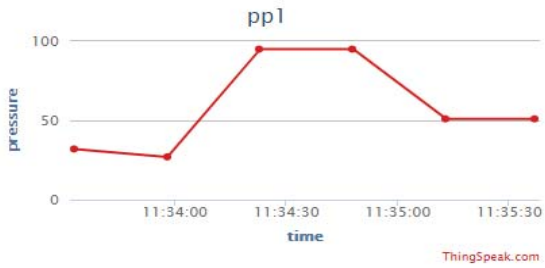

Fig. 6. (a) pressure point 1

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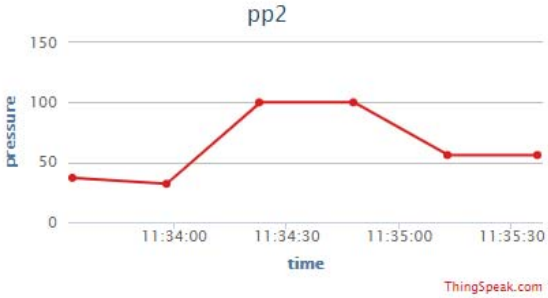

Fig. 7. Fig. 5(b) pressure point 2

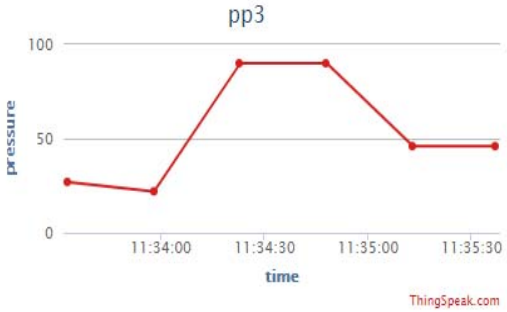

Fig. 8. Fig. 5(c) pressure point 3

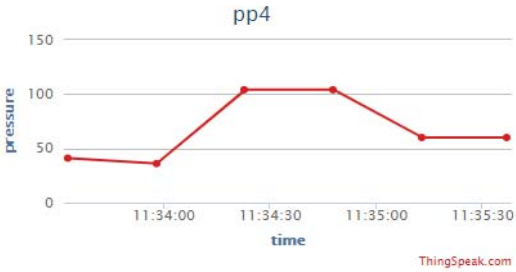

Fig. 9. Fig. 5(d) pressure point 4

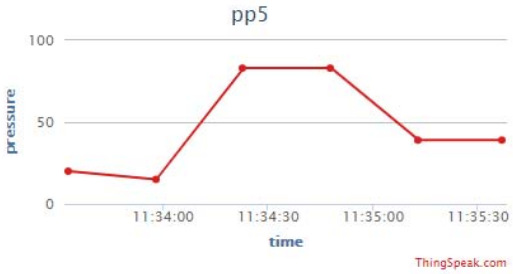

Fig. 10. Fig. 5(e) pressure point 5

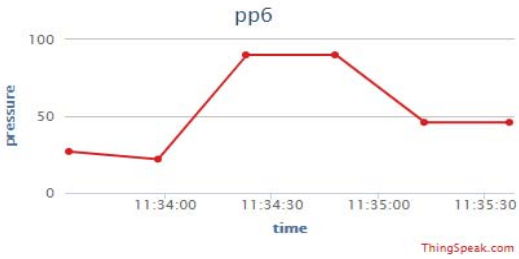

Fig. 11. Fig. 5(f) pressure point 6

Alert System

This system alerts the patient whenever there's an abnormality in the foot pressure, body temperature and pulse rate through android app called push bullet. Using matlab script over the cloud an alert system is designed to identify the unusual behavior of the patient and generate alerts by sending notification and messages to the patients and it concern doctor for monitoring. Figure 6 shows the system outcome as message and notification over the mobile.

- Our solution has represented how IOT can ease the doctor patient relation with such technological advancements. It provide sufficient data to analyze the patients medical conditions as well as stores the data for future use.
- The Healthcare kit allows patients to get more knowledge about the complication which arose because of diabetics

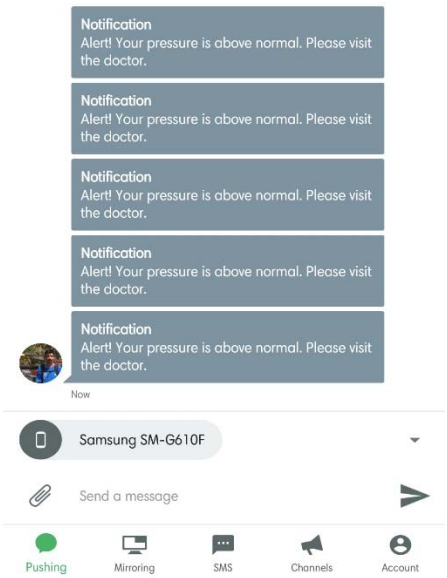

Fig. 12. Fig 6 message alerts

V. Conclusion

The optimal use of IOT to detect the foot ulcer in patient is boon for both the doctor and the patient. Leading to a better quality of life among the patients. The following system is an tried and tested development. Such deep study engender all kind of analysis with regards to every possible symptoms of foot ulcer. The use of open source ardware widens numerous opportunities for enhancement in the field of technology and data. The data helps the patients to detect the abnormality in heart and blood pressure

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